Ch 6 · Tricyclic antidepressant poisoning

Mechanisms

- Na-channel block (wide QRS, VT), alpha-block (hypotension), anticholinergic, GABA-A block (seizures)

ECG risk

- QRS > 100 ms seizures, > 160 ms arrhythmia; R in aVR > 3 mm

Treatment

- Sodium bicarbonate 1–2 mmol/kg repeated to pH 7.45–7.55; hyperventilate if intubated
- Benzodiazepines for seizures (no phenytoin); avoid physostigmine
- Noradrenaline; lidocaine for VT; charcoal if airway safe; lipid/ECMO for arrest

Key point: Bicarbonate is the antidote – titrate to QRS narrowing.

Ch 7 · Beta-blocker and calcium channel blocker poisoning

Features

- Bradycardia, AV block, hypotension; CCB: hyperglycaemia; BB: hypoglycaemia, bronchospasm
- Propranolol: seizures, wide QRS; sotalol: long QT

Treatment

- Charcoal; whole bowel irrigation for slow-release
- IV calcium; high-dose insulin 1 U/kg then 1–10 U/kg/h with glucose
- Noradrenaline/adrenaline, pacing; lipid, methylene blue, VA-ECMO; glucagon for BB

Key point: Hyperglycaemia severity tracks CCB poisoning severity.

Ch 8 · Sympathomimetic poisoning

Agents

- Cocaine, amphetamines, MDMA, cathinones, synthetic cannabinoids
- MDMA: hyponatraemia from ADH + water intake

Management

- Benzodiazepines generously; rapid cooling (ice water) to < 39 °C
- Hypertension: phentolamine/nitrates; avoid pure beta blockers with cocaine
- 3% saline for symptomatic hyponatraemia; fluids for rhabdomyolysis

Complications

- ACS, stroke, dissection, AKI, DIC, compartment syndrome

Key point: Hyperthermia kills – cool first, antipyretics do not work.

Ch 9 · Ethylene glycol and toxic alcohols

Labs

- High anion gap acidosis + high osmolar gap; oxalate crystals

Toxicity

- ADH → glycolate (acidosis) → oxalate (AKI, low Ca); methanol → formate (blindness)

Treatment

- Fomepizole (or ethanol)
- Haemodialysis: pH < 7.25, AKI, high levels, visual/neuro signs (EXTRIP)
- Bicarbonate; thiamine + pyridoxine (EG), folate (methanol)

Key point: Osmolar gap falls as the toxic acids rise – a normal gap late does not exclude it.

Ch 103 · SJS / TEN

Classification

- SJS < 10% BSA; overlap 10–30%; TEN > 30%
- Drugs: allopurinol, carbamazepine, lamotrigine, phenytoin, sulfonamides, nevirapine

SCORTEN

- Age > 40, malignancy, HR > 120, BSA > 10%, urea > 10, HCO3 < 20, glucose > 14

Management

- Stop drug; burns unit; warm room, fluids, nutrition, analgesia
- Non-adherent dressings, eye care (ophthalmology), mucosal care
- Immunomodulation (ciclosporin, etanercept, IVIG) – uncertain evidence

Key point: Stopping the culprit drug early is the most effective treatment.

Ch 104 · Alcohol withdrawal

Timeline

- 6–24 h tremor; 6–48 h seizures; 12–48 h hallucinosis; 48–96 h delirium tremens

Treatment

- Benzodiazepines: symptom-triggered (CIWA-Ar) or front-loading
- IV thiamine before glucose; Mg, K, PO4, fluids
- Refractory: phenobarbital, propofol; dexmedetomidine adjunct only

Wernicke's

- Confusion, ophthalmoplegia, ataxia; high-dose IV thiamine; untreated → Korsakoff

Key point: Antipsychotics alone lower seizure threshold – benzodiazepines first.